

Particle Production in de Sitter Spacetime

Marcell Howard

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Introduction

Here we derive the gravitational particle production in a de Sitter spacetime at 0-th order in the self-coupling parameter λ using the in-in formalism. We are interested in eventually going beyond 0-th order, so getting a handle on the simplest case can act as a guide for the more difficult calculation. This can be viewed as a warm-up to calculating the particle production at higher and higher powers of the self-coupling.

Conventions We use the mostly plus metric signature, i.e. $\eta_{\mu\nu} = (-, +, +, +)$ and units where $c = \hbar = 1$. The reduced four dimensional Planck mass is $M_{\text{Pl}} = (8\pi G)^{-1/2} \approx 2.43 \times 10^{18} \text{ GeV}$. We use boldface letters $\mathbf{r}$ to indicate 3-vectors and x and p to denote 4-vectors. Conventions for the curvature tensors, covariant and Lie derivatives are all taken from Carroll.

The Mode Functions in de Sitter Spacetime

We start with the action for a free scalar field

$$S = -\frac{1}{2} \int d^4x \sqrt{-g} (g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi + m^2 \phi^2 + \xi R \phi^2), \quad (1)$$

where the metric is given by

$$ds^2 = -dt^2 + a^2(t) \delta_{ij} dr^i dr^j, \quad a(t) = e^{H_I t}. \quad (2)$$

As typical, we shall work in conformal time $dt = a(\tau) d\tau$, bringing the spacetime interval to the form

$$ds^2 = a(\tau)(-d\tau^2 + \delta_{ij} dr^i dr^j), \quad d\tau = \frac{dt}{a(t)} = e^{-H_I t} \Rightarrow \tau(t) = -\frac{1}{H_I} e^{-H_I t} \Leftrightarrow a(\tau) = -\frac{1}{H_I \tau}. \quad (3)$$

The relation between the conformal time and the physical time follows from integrating both sides of the equation. Note, the conformal time takes on values in the interval $\tau \in (-\infty, 0)$. We have

$$a'(\tau) = \frac{1}{H_I \tau^2}, \quad a''(\tau) = -\frac{2}{H_I \tau^3}. \quad (4)$$

The Klein-Gordon equation for a scalar field in conformal FRW coordinates is

$$\phi''(\mathbf{r}, \tau) + 2\mathcal{H}\phi'(\mathbf{r}, \tau) - \nabla^2 \phi(\mathbf{r}, \tau) + a^2(m^2 + \xi R)\phi(\mathbf{r}, \tau) = 0, \quad (5)$$

where $\mathcal{H}(\tau) = \frac{a'}{a}$ is the conformal expansion rate. We can use a conformal scaling of the scalar field $\varphi = \frac{1}{a}\phi$ to get a more familiar version for the equations of motion

$$\varphi'' - \nabla^2 \varphi + a^2 \left[m^2 - \frac{a''}{a^3} (1 - 6\xi) \right] \varphi = 0. \quad (6)$$

The canonical quantization relations imposes a condition on the mode functions which we will write down here. This can be done by writing the general solution to the equations of motion and then quantizing the Fourier modes i.e.

$$\hat{\varphi}(\mathbf{r}, \tau) = \frac{1}{\sqrt{\mathcal{V}}} \sum_{\mathbf{p}} [\hat{a}_{\mathbf{p}} g_{\mathbf{p}}(\tau) e^{-i\mathbf{p} \cdot \mathbf{r}} + \hat{a}_{\mathbf{p}}^\dagger g_{\mathbf{p}}^*(\tau) e^{i\mathbf{p} \cdot \mathbf{r}}], \quad (7)$$

where the creation and annihilation operators are defined by

$$\hat{a}_{\mathbf{p}} |0\rangle = 0, \quad \langle 0| \hat{a}_{\mathbf{p}}^\dagger = 0, \quad (8)$$

with the state initially prepared in the Bunch-Davies vacuum and we quantize the field within a box of volume $\mathcal{V}$. We can write down the commutation relations for the field and its conjugate partner. First we write down the time derivative

$$\hat{\varphi}'(\mathbf{r}, \tau) = \frac{1}{\sqrt{\mathcal{V}}} \sum_{\mathbf{p}} [\hat{a}_{\mathbf{p}} g_p'(\tau) e^{-i\mathbf{p}\cdot\mathbf{r}} + \hat{a}_{\mathbf{p}}^\dagger g_p^{*'}(\tau) e^{i\mathbf{p}\cdot\mathbf{r}}], \quad (9)$$

and impose the familiar equal-time commutation relations

$$[\hat{\varphi}(\mathbf{r}, \tau), \hat{\varphi}'(\mathbf{r}', \tau)] = i\delta(\mathbf{r} - \mathbf{r}'), \quad (10)$$

which gives us

$$\begin{aligned} [\hat{\varphi}(\mathbf{r}, \tau), \hat{\varphi}'(\mathbf{r}', \tau)] &= \frac{1}{\mathcal{V}} \sum_{\mathbf{p}, \mathbf{q}} \left[[\hat{a}_{\mathbf{p}} g_p e^{-i\mathbf{p}\cdot\mathbf{r}}, \hat{a}_{\mathbf{q}} g_q' e^{-i\mathbf{q}\cdot\mathbf{r}'}] + [\hat{a}_{\mathbf{p}} g_p e^{-i\mathbf{p}\cdot\mathbf{r}}, \hat{a}_{\mathbf{q}}^\dagger g_q^{*'} e^{i\mathbf{q}\cdot\mathbf{r}'}] \right. \\ &\quad \left. + [\hat{a}_{\mathbf{p}}^\dagger g_p^* e^{i\mathbf{p}\cdot\mathbf{r}}, \hat{a}_{\mathbf{q}} g_q' e^{-i\mathbf{q}\cdot\mathbf{r}'}] + [\hat{a}_{\mathbf{p}}^\dagger g_p^* e^{i\mathbf{p}\cdot\mathbf{r}}, \hat{a}_{\mathbf{q}}^\dagger g_q^{*'} e^{i\mathbf{q}\cdot\mathbf{r}'}] \right], \end{aligned} \quad (11)$$

where we use the continuum limit for the summation i.e. $\frac{1}{\mathcal{V}} \sum \rightarrow \int \frac{d^3p}{(2\pi)^3}$. Using the usual commutation relations on the creation and annihilation operators

$$[\hat{a}_{\mathbf{p}}, \hat{a}_{\mathbf{q}}] = [\hat{a}_{\mathbf{p}}^\dagger, \hat{a}_{\mathbf{q}}^\dagger] = 0, \quad [\hat{a}_{\mathbf{p}}, \hat{a}_{\mathbf{q}}^\dagger] = (2\pi)^3 \delta^{(3)}(\mathbf{p} - \mathbf{q}), \quad (12)$$

the expression reduces down to

$$[\hat{\varphi}(\mathbf{r}, \tau), \hat{\varphi}'(\mathbf{r}', \tau)] = \int \frac{d^3p}{(2\pi)^3} \int \frac{d^3q}{(2\pi)^3} [g_p g_q^{*'} e^{-i(\mathbf{p}\cdot\mathbf{r} - \mathbf{q}\cdot\mathbf{r}')} \delta^{(3)}(\mathbf{p} - \mathbf{q}) - g_p^* g_q' e^{i(\mathbf{p}\cdot\mathbf{r} - \mathbf{q}\cdot\mathbf{r}')} \delta^{(3)}(\mathbf{q} - \mathbf{p})] \quad (13)$$

$$= \int \frac{d^3p}{(2\pi)^3} [g_p g_p^{*'} e^{-i\mathbf{p}\cdot(\mathbf{r} - \mathbf{r}')} - g_p^* g_p' e^{i\mathbf{p}\cdot(\mathbf{r} - \mathbf{r}')}]. \quad (14)$$

Recall that the mode functions $g_p(\tau)$ are invariant under $\mathbf{p} \rightarrow -\mathbf{p}$. Thus we can transform the integrand by $\mathbf{p} \rightarrow -\mathbf{p}$ to get

$$[\hat{\varphi}(\mathbf{r}, \tau), \hat{\varphi}'(\mathbf{r}', \tau)] = \int \frac{d^3p}{(2\pi)^3} [g_p g_p^{*'} - g_p^* g_p'] e^{i\mathbf{p}\cdot(\mathbf{r} - \mathbf{r}')} \quad (15)$$

Thus in order to ensure the usual commutation relations, we need the mode functions to be normalized such that

$$g_p(\tau) g_p^{*'}(\tau) - g_p^*(\tau) g_p'(\tau) = i. \quad (16)$$

Let's go back to the general expression for the conformally-scaled scalar field

$$\varphi(\mathbf{r}, \tau) = \int \frac{d^3p}{(2\pi)^3} [g_p(\tau) e^{i\mathbf{p}\cdot\mathbf{r}} \hat{a}_{\mathbf{p}} + g_p^*(\tau) e^{-i\mathbf{p}\cdot\mathbf{r}} \hat{a}_{\mathbf{p}}^\dagger], \quad (17)$$

which brings the equation of motion to the form

$$g_p''(\tau) + \left[p^2 + a^2 m^2 - \frac{a''}{a} (1 - 6\xi) \right] g_p(\tau) = 0. \quad (18)$$

The above already kind of looks like a Bessel equation

$$x^2 \frac{d^2 Y}{dx^2} + x \frac{dY}{dx} + (x^2 - \nu^2) Y(x) = 0, \quad (19)$$

so we'll make ansatz to bring the equation of motion more to the form that's required.

We'll take the ansatz to be

$$\begin{aligned} g_p &= (-\tau)^{1/2} f_p, \quad g_p' = (-\tau)^{1/2} f_p' - \frac{1}{2} (-\tau)^{-1/2} f_p, \\ g_p'' &= (-\tau)^{1/2} f_p'' - (-\tau)^{-1/2} f_p' - \frac{1}{4} (-\tau)^{-3/2} f_p, \end{aligned} \quad (20)$$

which results in the equation of motion becoming

$$f_p''(\tau) + \frac{1}{\tau} f_p'(\tau) + \left[p^2 - \frac{1}{\tau^2} \left(2(1 - 6\xi) + \frac{1}{4} - \frac{m^2}{H_I^2} \right) \right] f_p(\tau) = 0. \quad (21)$$

We can switch to a dimensionless variable $\bar{\tau} = -p\tau$ which brings the ODE to the form

$$\frac{d^2 f_p}{d\bar{\tau}^2} + \frac{1}{\bar{\tau}} \frac{df_p}{d\bar{\tau}} + \left[1 - \frac{\nu^2}{\bar{\tau}^2} \right] f_p = 0, \quad (22)$$

where

$$\nu^2 = 2(1 - 6\xi) + \frac{1}{4} - \frac{m^2}{H_I^2}. \quad (23)$$

The solution can be written down as a linear combination of Bessel functions. In this case, we expect complex solutions¹ and therefore choose to write down the Hankel functions

$$f_p(\tau) = A_p H_\nu^{(1)}(-p\tau) + B_p H_\nu^{(2)}(-p\tau), \quad (24)$$

¹We require complex solutions in order to satisfy the Wronskian condition.

where the Hankel functions are related to the usual Bessel functions by

$$H_\nu^{(1)}(z) = J_\nu(z) + iY_\nu(z), \quad H_\nu^{(2)}(z) = J_\nu(z) - iY_\nu(z). \quad (25)$$

We need to fix the integration constants using the boundary conditions. In order to fix the integration constants, we recognize that we want our mode functions at infinity to resemble the mode functions of Minkowski space

$$f_p \rightarrow \frac{e^{-ip\tau}}{\sqrt{2p}}. \quad (26)$$

The asymptotic behavior of both functions as $z \rightarrow -\infty$ are

$$H_\nu^{(1)}(z) \xrightarrow{z \rightarrow -\infty} \sqrt{\frac{2}{\pi z}} e^{i(z - \frac{\nu\pi}{2} - \frac{\pi}{4})}, \quad -\pi < \arg z < 2\pi, \quad (27)$$

$$H_\nu^{(2)}(z) \xrightarrow{z \rightarrow \infty} \sqrt{\frac{2}{\pi z}} e^{-i(z - \frac{\nu\pi}{2} - \frac{\pi}{4})}, \quad -2\pi < \arg z < \pi. \quad (28)$$

Thus, we have

$$g_p = \sqrt{-\tau} (A_p H_\nu^{(1)}(-p\tau) + B_p H_\nu^{(2)}(-p\tau)) \xrightarrow{\tau \rightarrow -\infty} \sqrt{-\tau} \left[A_p \sqrt{\frac{2}{\pi}} e^{-ip\tau} e^{-\frac{i\pi}{2}(\nu - \frac{1}{2})} + B_p \sqrt{\frac{2}{\pi}} e^{ip\tau} e^{\frac{i\pi}{2}(\nu + \frac{1}{2})} \right] \quad (29)$$

The matching condition of the asymptotic behavior implies $B_p = 0$. Therefore, we are left with

$$g_p(\tau) = \frac{\sqrt{\pi}}{2} \sqrt{-\tau} H_\nu^{(1)}(-p\tau) e^{\frac{i\pi}{2}(\nu + 1/2)}. \quad (30)$$

Going forward, we're going to assume that the scalar field is light compared to the scale of inflation ($\frac{m^2}{H_I^2} \ll 1$). We can specialize to the minimal coupling ($\xi = 0$) and conformally coupled ($\xi = 1/6$)

$$g_p(\tau) = \frac{\sqrt{-\pi\tau}}{2} \begin{cases} -H_{3/2}^{(1)}(-p\tau), & \xi = 0 \\ iH_{1/2}^{(1)}(-p\tau), & \xi = 1/6 \end{cases} \quad (31)$$

It would also be prudent to introduce a factor of $\sqrt{2p}$ into our definition of $g_p^<$

$$g_p(\tau) = \frac{1}{2p} \begin{cases} -\sqrt{\frac{\pi}{2}}\sqrt{-p\tau}H_{3/2}^{(1)}(-p\tau), & \xi = 0 \\ i\sqrt{\frac{\pi}{2}}\sqrt{-p\tau}H_{1/2}^{(1)}(-p\tau), & \xi = 1/6 \end{cases} \quad (32)$$

which is convenient because of the asymptotic behavior of the Hankel functions

$$H_\nu^{(1)} \xrightarrow{-p\tau \rightarrow \infty} \sqrt{\frac{2}{\pi}} \frac{e^{-ip\tau}}{\sqrt{-p\tau}} e^{\frac{-i\pi}{2}(\nu+1/2)}. \quad (33)$$

Recognizing the relationship between the Hankel functions and the spherical Bessel functions being

$$H_{\nu+1/2}^{(1)}(z) = \sqrt{\frac{2z}{\pi}}(j_\nu(z) + iy_\nu(z)), \quad (34)$$

with

$$j_1(z) = \frac{\sin z}{z^2} - \frac{\cos z}{z}, \quad y_1(z) = j_{-2}(z) = -\frac{\cos z}{z^2} - \frac{\sin z}{z}, \quad (35)$$

and therefore

$$j_1(z) + iy_1(z) = \frac{\sin z}{z^2} - \frac{\cos z}{z} - i\left(\frac{\cos z}{z^2} + \frac{\sin z}{z}\right) = -\frac{e^{iz}}{z} - \frac{ie^{iz}}{z^2} = -\frac{e^{iz}}{z}\left(1 + \frac{i}{z}\right). \quad (36)$$

This has the action of simplifying the mode functions to be

$$g_p(\tau) = \frac{e^{-ip\tau}}{\sqrt{2p}} \begin{cases} 1 - \frac{i}{p\tau}, & \xi = 0 \\ 1, & \xi = 1/6 \end{cases} \quad (37)$$

We can see here that a light² conformally-coupled scalar field has mode functions that are identical to those of a massless field in Minkowski space. From here, it is obvious that there *can't* be much particle production for these types of particle candidates. We can see the say that a light conformally-coupled scalar simply expands with the background spacetime and therefore the universe appears to be Minkowski for these particles.

²At least when compared to the scale of inflation.

The Power Spectrum

Now we'll specialize to the minimal coupling case $\xi = 0$. Let's calculate the equal-time real-space correlation function given by

$$\lim_{\tau \rightarrow 0} \langle \phi(\mathbf{r}, \tau) \phi(\mathbf{r}', \tau) \rangle. \quad (38)$$

Since the full expression for ϕ is given by

$$\phi(\mathbf{r}, \tau) = \frac{1}{a} \int \frac{d^3 p}{(2\pi)^3} [g_p(\tau) e^{i\mathbf{p} \cdot \mathbf{r}} \hat{a}_{\mathbf{p}} + g_p^*(\tau) e^{-i\mathbf{p} \cdot \mathbf{r}} \hat{a}_{\mathbf{p}}^\dagger], \quad g_p(\tau) = \frac{e^{-ip\tau}}{\sqrt{2p}} \left(1 - \frac{i}{p\tau} \right). \quad (39)$$

Given that the annihilation operator is defined by $\hat{a}_{\mathbf{p}} |0\rangle$, the commutation relation implies that

$$\langle \hat{a}_{\mathbf{p}} \hat{a}_{\mathbf{p}}^\dagger \rangle = \langle 0 | \hat{a}_{\mathbf{p}} \hat{a}_{\mathbf{q}}^\dagger | 0 \rangle = (2\pi)^3 \delta^{(3)}(\mathbf{p} - \mathbf{q}), \quad (40)$$

and this is the only term that contributes. Therefore, the correlation function becomes

$$\begin{aligned} \langle \phi(\mathbf{r}, \tau) \phi(\mathbf{r}', \tau) \rangle &= \frac{1}{a^2} \int \frac{d^3 p}{(2\pi)^3} \int \frac{d^3 q}{(2\pi)^3} g_p(\tau) g_q^*(\tau) e^{-i\mathbf{p} \cdot \mathbf{r}} e^{i\mathbf{q} \cdot \mathbf{r}'} \langle \hat{a}_{\mathbf{p}} \hat{a}_{\mathbf{p}}^\dagger \rangle \\ &= \frac{1}{a^2} \int \frac{d^3 p}{(2\pi)^3} |g_p(\tau)|^2 e^{-i\mathbf{p} \cdot (\mathbf{r} - \mathbf{r}')}. \end{aligned} \quad (41)$$

Given that the universe is translationally invariant, we can set $\mathbf{r}' = 0$ without any loss of generality, bringing the correlation function to the form of

$$\langle \phi(\mathbf{r}, \tau) \phi(0, \tau) \rangle = \frac{1}{a^2} \int \frac{d^3 p}{(2\pi)^3} |g_p(\tau)|^2 e^{-i\mathbf{p} \cdot \mathbf{r}} = \frac{1}{2\pi^2 a^2(\tau)} \int_0^\infty p^3 |g_p(\tau)|^2 \frac{dp}{p}, \quad (42)$$

where we introduce a new function, $\mathcal{P}(p, \tau)$, called the power spectrum.

$$\mathcal{P}(p, \tau) = \frac{p^3}{2\pi^2 a^2(\tau)} |g_p(\tau)|^2. \quad (43)$$

We choose to align our coordinate system along the z -axis³. This brings the correlation function to the form

³Translational-invariance implies we can orient our coordinate system anywhere in space and the result remains the same.

$$\langle \phi(\mathbf{r}, \tau) \phi(0, \tau) \rangle = \frac{1}{a^2} \frac{1}{(2\pi)^2} \int_0^\infty \int_{-1}^1 p^2 |g_p|^2 e^{-ipr \cos \theta} d \cos \theta dp = \frac{1}{a^2} \frac{1}{(2\pi)^2} \int_0^\infty p^2 |g_p|^2 \frac{e^{-ipr \cos \theta}}{-ipr} \Big|_{\cos \theta = -1}^{\cos \theta = 1}, \quad (44)$$

which yields

$$\langle \phi(\mathbf{r}, \tau) \phi(0, \tau) \rangle = \frac{1}{a^2} \frac{1}{(2\pi)^2} \int_0^\infty \left(1 + \frac{1}{(p\tau)^2} \right) \frac{\sin pr}{r} dp = \frac{H_I^2}{(2\pi)^2} \int_0^\infty \left(\frac{1}{\bar{p}^2} + \frac{\tau^2}{r^2} \right) \sin \bar{p} d\bar{p}, \quad (45)$$

where $\bar{p} = pr$. If we were to take $\tau \rightarrow 0$ limit,

$$\lim_{\tau \rightarrow 0} \langle \phi(\mathbf{r}, \tau) \phi(0, \tau) \rangle = \frac{H_I^2}{(2\pi)^2} \int_0^\infty \frac{\sin \bar{p}}{\bar{p}^2} d\bar{p}, \quad (46)$$

it's clear that the correlation function is divergent in the IR. The physical interpretation can be reasoned from the following: the IR limit corresponds to low momenta which is equivalent to large distances. However, a de Sitter spacetime, because of its positive cosmological constant, gets larger without end⁴. Therefore, the correlation function is sampling points within a volume that only ever grows larger and can't converge as a result. However, if we take the *coincident limit* i.e. the limit where $\mathbf{r}' \rightarrow \mathbf{r}$, the correlation function takes on a very simple form

$$\langle \phi^2(\mathbf{r}, \tau) \rangle = \frac{H_I^2}{(2\pi)^2}. \quad (47)$$

It should be noted that this result only comes about because we neglected the mass of the field. Starting from the mode functions of the canonical field

$$\phi_p(\tau) = \frac{\chi_p(\tau)}{a(\tau)} = -\frac{\sqrt{\pi}}{2} H_I(-\tau)^{3/2} H_\nu^{(1)}(-p\tau), \quad (48)$$

with ν being given by

$$\nu = \sqrt{\frac{9}{4} - \frac{m^2}{H_I^2}} = \frac{3}{2} \sqrt{1 - \frac{4m^2}{9H_I^2}} \simeq \frac{3}{2} \left(1 - \frac{2m^2}{9H_I^2} \right) = \frac{3}{2} - \frac{m^2}{3H_I^2} \equiv \frac{3}{2} - u, \quad (49)$$

and we've defined a new parameter

$$u = \frac{m^2}{3H_I^2} \ll 1. \quad (50)$$

⁴The jargon is that de Sitter is eternal.

Now we are in a position to compute the correlation function:

$$\langle \phi(\mathbf{r}, \tau) \phi(\mathbf{r}', \tau') \rangle \equiv \langle 0 | \phi(\mathbf{r}, \tau) \phi(\mathbf{r}', \tau') | 0 \rangle \quad (51)$$

$$= \left\langle 0 \left| \int \frac{d^3 p}{(2\pi)^3} \int \frac{d^3 q}{(2\pi)^3} [\phi_p(\tau) e^{i\mathbf{p} \cdot \mathbf{r}} \hat{a}_{\mathbf{p}} + \phi_p^*(\tau) e^{-i\mathbf{p} \cdot \mathbf{r}} \hat{a}_{\mathbf{p}}^\dagger] [\phi_q(\tau') e^{i\mathbf{q} \cdot \mathbf{r}'} \hat{a}_{\mathbf{q}} + \phi_q^*(\tau') e^{-i\mathbf{q} \cdot \mathbf{r}'} \hat{a}_{\mathbf{q}}^\dagger] \right| 0 \right\rangle \quad (52)$$

$$= \int \frac{d^3 p}{(2\pi)^3} \int \frac{d^3 q}{(2\pi)^3} \phi_p \phi_q^* e^{i\mathbf{p} \cdot \mathbf{r}} e^{-i\mathbf{q} \cdot \mathbf{r}'} \langle 0 | \hat{a}_{\mathbf{p}} \hat{a}_{\mathbf{q}}^\dagger | 0 \rangle \quad (53)$$

$$= \int \frac{d^3 p}{(2\pi)^3} \phi_p(\tau) \phi_p^*(\tau') e^{i\mathbf{p} \cdot (\mathbf{r} - \mathbf{r}')}. \quad (54)$$

Introducing a new variable $R = |\mathbf{r} - \mathbf{r}'|$, and the angle between to be $\mathbf{r} \cdot \mathbf{r}' = \cos \theta$.

This gives

$$\langle \phi(\mathbf{r}, \tau) \phi(\mathbf{r}', \tau') \rangle = \int \frac{d^3 p}{(2\pi)^3} \phi_p(\tau) \phi_p^*(\tau') e^{ipR \cos \theta}. \quad (55)$$

We can break the 3D integral up by

$$\langle \phi(\mathbf{r}, \tau) \phi(\mathbf{r}', \tau') \rangle = \frac{1}{2\pi^2} \int_0^\infty p^2 \frac{\sin pR}{pR} \phi_p(\tau) \phi_p^*(\tau') dp. \quad (56)$$

Now we shall implement the *long wavelength approximation*:

$$\frac{p/a(t)}{H(t)} = -p\tau < \epsilon \ll 1, \quad (57)$$

and similarly for the prime coordinates. We've introduced a small parameter ϵ . In this small wavelength approximation, the mode functions become

$$\phi_p(\tau) = -\frac{\sqrt{\pi}}{2} H_I(-\tau)^{3/2} H_\nu^{(1)}(-p\tau) = -\frac{\sqrt{\pi}}{2} H_I(-\tau)^{3/2} [J_\nu(-p\tau) + iY_\nu(-p\tau)]. \quad (58)$$

We can approximate the Bessel functions as

$$J_\nu(-p\tau) = \sum_{m=0}^{\infty} \frac{(-1)^m}{m! \Gamma(m + \nu + 1)} \left(\frac{-p\tau}{2} \right)^{2m+\nu} \simeq \frac{1}{\Gamma(\nu + 1)} \left(\frac{-p\tau}{2} \right)^\nu = \frac{4}{3\sqrt{\pi}} \left(\frac{-p\tau}{2} \right)^{3/2}, \quad (59)$$

$$Y_\nu(-p\tau) = \frac{J_\nu(-p\tau) \cos \pi\nu - J_{-\nu}(-p\tau)}{\sin \pi\nu} \simeq -\frac{1}{2\sqrt{\pi}} \left(\frac{-p\tau}{2} \right)^{-3/2+\nu}. \quad (60)$$

Since $-p\tau \sim \epsilon$, terms that look like $J_\nu \sim (-p\tau)^\nu$ can be neglected and $J_{-\nu} \sim (-p\tau)^{-\nu}$ must be kept. Therefore, we can approximate the mode functions as

$$\phi_p(\tau) \simeq -\frac{\sqrt{\pi}}{2}(-\tau)^{3/2} \cdot \frac{iH_I}{-2\sqrt{\pi}} \left(\frac{-p\tau}{2} \right)^{-3/2+u} = \frac{iH_I}{2^2} \left(\frac{2}{p} \right)^{3/2} (-p\tau)^u 2^{-u} \simeq \frac{iH_I}{\sqrt{2p^3}} (-p\tau)^u. \quad (61)$$

The long wavelength approximation imposes a cutoff to the integral

$$\langle \phi(\mathbf{r}, \tau) \phi(\mathbf{r}', \tau') \rangle_{\lambda^0, L} = \frac{1}{2\pi^2} \int_0^{-\epsilon/\tau_{\min}} p^2 \frac{\sin pR}{pR} \left(\frac{iH_I}{\sqrt{2p^3}} (-p\tau)^u \right) \left(\frac{-iH_I}{\sqrt{2p^3}} (-p\tau')^u \right) dp \quad (62)$$

$$= \frac{H_I^2}{(2\pi)^2} (\tau\tau')^u \int_0^{-\epsilon/\tau_{\min}} \frac{dp}{p} \frac{\sin pR}{pR} p^{2u}, \quad (63)$$

where $\tau_{\min} \equiv \min(\tau, \tau')$ and the subscript λ^0 and L denotes the zero-th order in the coupling taking the long wavelength approximation. Now we take the coincident limit $\mathbf{r}' = \mathbf{r}$ and $\tau' = \tau$ i.e. $r \rightarrow 0$ limit. Recall

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1. \quad (64)$$

The correlation function in the coincident limit becomes

$$\langle \phi^2(\mathbf{r}, \tau) \rangle_{\lambda^0, L} = \lim_{R \rightarrow 0} \frac{H_I^2}{(2\pi)^2} (-\tau)^{2u} \int_0^{-\epsilon/\tau_{\min}} \frac{dp}{p} \frac{\sin pR}{pR} p^{2u} = \frac{H_I^2}{(2\pi)^2} (-\tau)^{2u} \int_0^{-\epsilon/\tau_{\min}} \frac{dp}{p^{1-2u}} \quad (65)$$

$$= \frac{H_I^2}{8\pi^2} \frac{(-\tau_{\min})^{2u}}{u} \left(\frac{\epsilon}{-\tau_{\min}} \right)^{2u} = \frac{H_I^2}{8\pi^2} \frac{\epsilon^{2u}}{u} \simeq \frac{H_I^2}{8\pi^2} \frac{1}{u} = \frac{3H_I^4}{8\pi^2 m^2}, \quad (66)$$

where we used $\epsilon^{2u} \approx 1 + 2u \log \epsilon \simeq 1$ since u is small. We can see that the correlation function for the full field in de Sitter spacetime is just a constant.